

CIVIL, MARITIME AND ENVIRONMENTAL ENGINEERING
(CMEE)



Funded by
the European Union



TECHNICAL WORKSHOP

DATE

23 September 2026

TIME

8:30 - 15:30

PROJECT WEBSITE

revive.bridges2f.com



CONSORTIUM



THE PROJECT

HORIZON-MSCA-SE-2025

Non-Destructive Evaluation and Service Life Extension of Vulnerable Reinforced Concrete Infrastructure

The aim of REVIVE is to provide a knowledge exchange platform for academic researchers and engineers to develop advanced engineering tools that can efficiently and reliably assess and predict the current and future condition of RC structures.

It also seeks to pioneer next-generation sustainable methods for repair and retrofitting, focusing on extending service life through appropriate monitoring technologies.

THE PROBLEM

Inspection of RC structures still depends largely on manual assessment and subjective human judgement. Current practices are reactive, fragmented, and insufficiently predictive.

SCALE

300,000

concrete bridges in Europe,
valued at €300 billion

€8bn

annual maintenance cost

3,200+

UK bridges reported substandard
in 2022

8%

of global CO₂ emissions from
concrete

PROJECT OBJECTIVES

The overall aim is to advance the state of the art in NDE, SHM, predictive modelling, and sustainable and smart materials, and to integrate these into digitally enabled, performance-based asset management frameworks.

01 Develop and validate an integrated NDE–SHM methodology for RC infrastructure assessment

Five or more complementary techniques — GPR, ultrasonic, vibration, fibre optics — in one inspection framework

WPS 1–2 · M18

02 Quantify structural performance and vulnerability through predictive digital twin modelling

Physics-based and AI-enhanced models calibrated with multi-sensor inspection data

WP3 · M30

03 Validate sustainable service life extension strategies using low-carbon and smart materials

Recycled aggregate concretes and smart composites characterised in laboratory and field conditions

WP4 · M36

04 Integrate NDE, SHM and predictive modelling into asset management frameworks

Harmonised guidelines and digital tools for condition-based maintenance of bridges and tunnels

WP5 · M42

05 Promote intersectoral knowledge transfer and structured skills development through mobility

Secondments across academia, industry and certification partners, with joint workshops and training schools

WPS 6–7 · M1–48

06 Enhance visibility and uptake of REVIVE outcomes through open training and outreach

Summer schools, technical workshops and a hackathon, open to participants beyond the consortium

WPS 6–7 · M42

THE CONSORTIUM

12 partners · 9 countries

COORDINATOR **University of Southampton**
Civil, Maritime and Environmental Engineering · United Kingdom

BENEFICIARIES



University of Bristol

United Kingdom



TU Delft

Netherlands



University of Ljubljana

Slovenia



Università di Roma Tor Vergata

Italy



Helmut Schmidt University

Germany



HUN-REN SZTAKI

Hungary



ZAG

Slovenia

ASSOCIATED PARTNERS



SVTI

Switzerland



AtkinsRéalis

United Kingdom



Timezyx

Sweden



SHM Next

Netherlands

WORKSHOP AIMS

Open event

This technical workshop launches REVIVE as an open event. It brings together infrastructure owners, industry and researchers from inside and beyond the consortium to share current practice and current research on assessing ageing structures.

01 An open launch

Mark the start of REVIVE with an event open to the wider infrastructure and research community, not only to project partners.

02 Set the industry baseline

Keynotes from Network Rail, National Highways and AtkinsRéalis on the real challenges of managing ageing structures.

03 Hear a wide range of voices

Twelve talks from asset owners, industry, and universities and institutes across Europe, several from outside the consortium.

04 Cover the technical ground

NDE, structural health monitoring, digital twins, Bayesian methods, fibre optics, satellite monitoring and machine learning.

05 Build connections

Create room for discussion and new collaborations between researchers, asset owners and industry over the course of the day.

06 See the facilities

Close with lab tours of the testing and monitoring facilities at Southampton.

LAUNCHING THE BRIDGES TO THE FUTURE LAB

From lab to life

Bridges to the Future Lab (B2F)

University of Southampton

A new research lab dedicated to making bridges and critical infrastructure safer, smarter, more resilient and more sustainable.

B2F brings together experimental research, advanced computational modelling, non-destructive evaluation, structural monitoring, digital technologies and field implementation to address the challenges facing existing and future infrastructure.



bridges2f.com



B2F AND REVIVE: FROM RESEARCH TO REAL-WORLD IMPACT

THE LAB



A broader research platform focused on:

- Structural resilience and multi-hazard performance
- Ageing and deteriorating infrastructure
- Intelligent monitoring, sensing and digital twins
- Sustainable materials and bridge systems
- Assessment, maintenance, retrofit and service-life extension

THE PROJECT



Non-Destructive Evaluation and Service Life Extension of Vulnerable Reinforced Concrete Infrastructure

- Non-destructive evaluation
- Structural health monitoring
- Deterioration assessment
- Predictive modelling
- Repair, retrofit and service-life extension

OBSERVE → UNDERSTAND → PREDICT → INTERVENE → EXTEND LIFE

The REVIVE workshop marks both the beginning of a new international research collaboration and the launch of the Bridges to the Future Lab.

MORNING

8:30 – 11:00

TIME	ACTIVITY	SPEAKER
8:30 - 9:00	ARRIVAL AND REGISTRATION	
9:00 - 9:15	Welcome and workshop introduction by Head of CMEE	Prof Yannis Ieropoulos University of Southampton
9:15 - 9:30	Introduction to REVIVE and purpose of the workshop	Dr Mehdi Kashani University of Southampton
9:30 - 10:00	Keynote: Improved resilience by unlocking hidden truths	Mr Tim Kersley Network Rail
10:00 - 10:30	Keynote: Challenges in the management of ageing highway structures	Mr Colin George National Highways
10:30 - 11:00	Keynote: Advancements in NDT for bridges with hidden critical elements - Findings from National Highways Structures Moonshot	Dr Chris Mundell AtkinsRéalis

LATE MORNING

11:00 – 13:10

TIME	ACTIVITY	SPEAKER
11:00 - 11:30	COFFEE BREAK	
11:30 - 11:50	UK Collaboratorium for Research on Infrastructure and Cities (UKCRIC)	Prof William Powrie University of Southampton
11:50 - 12:10	Challenges in the structural assessment of transfer slabs for shear	Prof Robert Vullum Imperial College London
12:10 - 12:30	Application of high-resolution distributed optical fibre in damage assessment of ageing RC structures	Ms Saba Ghassemi University of Southampton
12:30 - 12:50	Reliability assessment of damage diagnosis with Bayesian updating: Current status and future directions	Dr Francesca Marsili Helmut-Schmidt University
12:50 - 13:10	Structural monitoring of existing bridges via satellite-based techniques	Dr Fabio Di Carlo Università di Roma Tor Vergata

AFTERNOON

13:10 – 15:30

TIME	ACTIVITY	SPEAKER
13:10 - 14:10	LUNCH	
14:10 - 14:30	Structural health monitoring with Bayesian inversion and probabilistic models: Tools, limitations, and challenges	Dr Noemi Friedman SZTAKI
14:30 - 14:50	Data-driven evaluation of existing bridges: Methods and applications at ZAG	Dr Andrej Anžlin ZAG
14:50 - 15:10	Smart tape for structural health monitoring	Dr Meisam Jalalvand University of Southampton
15:10 - 15:30	Real-time structural damage localisation using modal curvatures based on Machine Learning algorithm	Dr Masood Mohandes University of Southampton
15:30	Closing remarks and Lab tours	Dr Mehdi Kashani